245B: Sample midterm questions

Terence Tao · 25 January 2009

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Below the fold, I am giving some sample questions for the 245B midterm next week. These are drawn from my previous 245A and 245B exams (with some modifications), and the solutions can be found by searching my previous web pages for those courses. (The homework assignments are, of course, another good source of practice problems.) Note that the actual midterm questions are likely to be somewhat shorter than the ones provided here (this is particularly the case for those questions with multiple parts). More info on the midterm can be found at the class web page, of course.

(These questions are of course primarily intended for my students than for my regular blog readers; but anyone is welcome to comment if they wish.)

Question 1. Let $(X, \mathcal{X}, \mu)$ be a σ-finite measure space, and let $\nu : \mathcal{X} \to \mathbb{R}$ be a signed σ-finite measure. Show that the following are equivalent:

1. $-\mu \leq \nu \leq \mu$.
2. $|\nu| \leq \mu$.
3. $\nu \ll \mu$ and $\left|\frac{d\nu}{d\mu}\right| \leq 1$ μ-a.e.. $\diamond$

Question 2. Let $(\mathbb{R}, \mathcal{B})$ be the real line with the Borel σ-algebra, and let $\mu : \mathcal{B} \to [0, +\infty)$ be a σ-finite unsigned measure. Show that it is not possible for counting measure $\#$ on the real line (restricted to $\mathcal{B}$) to be absolutely continuous with respect to μ, i.e. $\# \not\ll \mu$. $\diamond$

Question 3. Let $1 \leq p < \infty$, let $\ell^p(\mathbb{N})$ be the Banach space of pth power summable real sequences $(a_n)_{n \in \mathbb{N}}$ with the usual ℓ^p norm. For each natural number n, let e_n be the element of $\ell^p(\mathbb{N})$ consisting of the sequence which equals 1 at the nth entry and 0 elsewhere, thus $(e_n)_m = 1$ when $n = m$ and $(e_n)_m = 0$ otherwise. Let $a_1, a_2, a_3, \ldots$ be a sequence in a Banach space X.

1. Show that there is at most one continuous linear transformation $T : \ell^p(\mathbb{N}) \to X$ with the property that $Te_n = a_n$ for all natural numbers n.
2. If $p = 1$, show that there exists a continuous linear transformation $T : \ell^1(\mathbb{N}) \to X$ with $Te_n = a_n$ for all natural numbers n if and only if the sequence $a_1, a_2, \ldots$ is bounded. $\diamond$
3. If $p = \infty$, show that the uniqueness claim 1. can fail. $\diamond$

Question 4. Let W be a vector space, let A be an index set, and for every $\alpha \in A$, let V_α be a subspace of W which is equipped with a norm $\| \|_{V_\alpha}$. Suppose that for each α, the normed vector space $(V_\alpha, \| \|_{V_\alpha})$ is a Banach space. Assume also the following compatibility condition: if $\alpha, \beta \in A$ and x_n is a sequence in $V_\alpha \cap V_\beta$ which converges in V_α norm to x and in V_β norm to y, then x is necessarily equal to y. Show that the space

$$V := \{x \in \bigcap_{\alpha \in A} V_\alpha : \sum_{\alpha \in A} \|x\|_{V_\alpha} < \infty\},$$

equipped with the norm

$$\|x\|_V = \sum_{\alpha \in A} \|x\|_{V_\alpha},$$

is also a Banach space. $\diamond$

Question 5. Let H be a Hilbert space, and let

$$V_1 \subset V_2 \subset V_3 \subset \ldots \subset H$$

be an increasing sequence of closed subspaces of H. Let $V := \overline{\bigcup_{n=1}^{\infty} V_n}$ be the closure of the union of these subspaces; this is another closed subspace of H. Show that for any x in H, the sequence $\pi_{V_n}(x)$ converges in norm to $\pi_V(x)$, where $\pi_V(x)$ is the orthogonal projection of x to V, and similarly for $\pi_{V_n}(x)$. ⋄

Question 6. Let H be a Hilbert space, and let V be a closed subspace of H which is non-trivial (i.e. not equal to $\{0\}$). Let π_V be the orthogonal projection onto V.

1. Show that π_V has operator norm 1.
2. Show that π_V is self-adjoint, i.e. $\pi_V^* = \pi_V$.
3. Show that π_V is idempotent, i.e. $\pi_V^2 = \pi_V$.
4. Conversely, if $P : H \to H$ is a self-adjoint idempotent continuous linear transformation of operator norm 1, show that $P = \pi_V$ for some non-trivial subspace of H. ⋄

[*Update*, Feb 2: Question 4 corrected.]

Comments (7)

Anonymous · 25 January 2009

What's this "ocunting measure" in question 2? Sounds dirty ... *[Fixed – T.]*

Anonymous · 30 January 2009

For question 2, I also don't understand the counting measure. So it lives on the sigma-algebra consisting of all the subsets of R, but niu is with Borel sigma-algebra. So they lives on different sigma-algebra.

Terence Tao · 30 January 2009

Fair enough; I intended to restrict counting measure to the Borel sigma algebra, and have now indicated this accordingly. (A measure on a sigma algebra can always be restricted to a sub-sigma algebra without difficulty.)

Anonymous · 15 April 2009

Dear professor!
I have a question which relate adjoint space. That is: " Is $C^*[0,1]$ linear isomorphism with $C[0,1]$?"
$C[0,1]$ is space of continous function in $[0,1]$. If we choose inner product of $C[0,1]$ is : $C[0,1] \longrightarrow \mathbb{R}$, by $= \int_0^1 f(x)g(x)dx$ then $C^*[0,1] = C[0,1]$ (isormophism)?
Thank you very much!

Adam Azzam · 5 February 2013

I hate to be this comment; but I suppose that V should also be non-trivial (so that the operator norm of the orthogonal projection is non-zero). *[Corrected, thanks – T.]*

Aditya Guha Roy · 29 December 2022

Wow, I really like this collection of problems.
Do you also have a sample final term, or is it possible to share the actualy finals and actual mid term questions? I followed this series of lectures very closely, and would surely like to give the problems a shot.
(I'd soon post my solutions / attempts to solve these problems in a comment below.)

Aditya Guha Roy · 29 December 2022

Solution to Q6:
Part 1: Follows from Pythagorean theorem, and by taking a unit vector in the subspace.

Part 2: We have $\langle \pi_V(x), y\rangle = \langle \pi_V(x), \pi_V(y)\rangle + \langle \pi_V(x), y - \pi_V(y)\rangle$ but the second term vanishes since the residual $y - \pi_V(y)$ is orthogonal to every element of V and hence in particular it is orthogonal to $\pi_V(x)$. Similarly, one has $\langle x, \pi_V(y)\rangle = \langle \pi_v(x), \pi_V(y)\rangle$.
Part 3: This follows from definition, since $\pi_v(y)$ is the unique element of $z \in V$ which minimizes $||z - x||$ over all $z \in V$.
Part 4: First of all notice that the range of a bounded idempotent operator is closed ($Px_n \to z \implies P(x_n) = P^2(x_n) \to P(z)$ thus $P(z) = z$ in particular z is in the range of P).
Let V be the range of P; then one has (thanks to idempotent property) $x = Px$ for all $x \in V$. Now we have $\langle x, y\rangle = \langle Px, y\rangle = \langle x, Py\rangle (thankstoself - adjointproperty), thusweget$latex \langle x , y – Py \rangle = 0$ for all $x \in V$; but this is equivalent to saying that $||y - Py|| = \inf_{z\in V}||y - z||$ or in other words, that Py is the orthogonal projection of y onto V. The claim follows. □
Solution to Q5: We will use the fact that projections are bounded linear operators. (One way to show this is to use Part 1 of Q6.)
To begin, notice that since V_n is a closed subspace of V so $\pi_{V_n} \circ \pi_V = \pi_{V_n}$. Thus it suffices to show that $\left|\left|\pi_{V_n}(y) - y\right|\right| \to 0$ for all $y \in V$.
If $y \in V$ then there is a sequence (y_n) of points $y_n \in V_n$ such that $||y_n - y|| \to 0$ as $n \to \infty$. One has (by the triangle inequality):
$\left|\left|\pi_{V_n}(y) - y\right|\right| \leq \left|\left|\pi_{V_n}(y_n) - y_n\right|\right| + \left|\left|\pi_{V_n}(y) - \pi_{V_n}(y_n)\right|\right| + ||y_n - y||$; using uniform boundedness principle the second term on the right hand side of the line of display goes to zero as $n \to \infty$, and the other quantities also go to zero as $n \to \infty$ since $y_n \to y$, as $n \to \infty$.
(PS: The comment above this one was made without reading that the solutions are already available; so I'll post the solutions, only if I find signficantly distinct solutions to the remaining ones'.)